# SOCANET：A DEEP LEARNING WETLAND CLASSIFICATION STRATEGY BASED ON IMPROVED CNN AND ATTENTION MECHANISM

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***Abstract*—Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in extracting local features for complex scene analysis. However, their ability to capture global feature representations is limited, posing challenges for complex land cover classification. To address these limitations, we propose a hybrid network structure named SOCANet, which integrates convolutional operations with linear self-attention mechanisms. In detail, the network is divided into encoder and decoder two parts. In the encoder, a cross-modal feature fusion module and a multi-level feature extraction module are integrated. The multi-level feature extraction module adopts a parallel structure and combines CNN and focused linear attention to improve the feature expression ability. The decoder part consists of two classifiers, which are responsible for different levels of feature classification tasks. Furthermore, the Nanjishan Wetland Nature Reserve region is used to verified the effective of our proposed method. Experimental results indicate that SOCANet achieves a comprehensive classification accuracy (OA) of 97% and a Kappa coefficient of 0.959, significantly outperforming mainstream baseline methods. Notably, the proposed approach excels in distinguishing complex land cover types, such as carex, mudflat, reeds, et al., offering a novel solution for feature learning and fusion in intricate scenarios.**

***Keywords—Multi-source remote sensing; SOCANet; Cross-modal feature fusion; Linear attention; Wetland mapping***

## I. INTRODUCTION

Wetland ecosystem plays a key role in global climate regulation, biodiversity conservation and ecological restoration, and the accurate classification of its surface features is crucial for ecological restoration, species monitoring and environmental protection [1]. In recent years, massive multi-source remote sensing data have provided more detailed feature information for the classification of wetland features, laying a solid foundation for the interpretation and application of remote sensing data [2]. However, how to efficiently process these multi-source data and extract more comprehensive and accurate information than single-source data has become an urgent technical problem in the field of automatic processing of remote sensing data [3].

With the rapid growth of computing power and data scale, deep learning has made significant progress in the field of computer vision with its powerful nonlinear feature extraction capability [4]. In general, the deep learning in semantic segmentation method involves the CNN-based and transformer-based. CNN and its variants have attracted much attention due to their accurate extraction of local features. However, the limitation of their local receptive field leads to limitations in capturing global features and remote feature dependencies [5]. The traditional transformer model has outstanding performance in context modeling, its insufficient efficiency in sequence extension limits its application in high-resolution remote sensing image analysis [6,7].

In order to solve the above problems, this paper proposes a hybrid deep learning model, SOCANet, which combines J-CNN and linear self-attention, aiming to achieve the fine classification of wetland surface objects through cross-modal feature fusion and multi-level feature extraction.

The main contributions of this paper are as follows:

1) SOCANet was compared with SOTA algorithms, including RMT [8], CoAtNet [9], Conformer [10] on our homemade dataset. The proposed method outperforms existing SOTA approaches.

2) In our model, the variable convolution of the edge is added to enhance the extraction of edge information, which solves the problem that the wetland boundary is difficult to distinguish. The linear attention module is added to enhance the expression ability of self-attention, which solves the problem of high similarity and easy misclassification of wetland features. A feature fusion module is added to solve the problem of cross-modal feature fusion mismatch caused by different imaging principles between optical image and SAR image.

## II. Study Area and Data Set

### A. Study area

This study focuses on the Nanjishan Wetland Nature Reserve, located in the southern part of Poyang Lake, Jiangxi Province, China. As the largest freshwater lake in China, Poyang Lake plays a crucial role in water resource regulation, maintaining ecosystem balance, and conserving biodiversity. It also serves as an ideal region for studying wetland ecosystems. The geographical extent of the study area is between 28°22'–29°45' N latitude and 115°47'–116°45' E longitude, covering a total area of approximately 162,200 square kilometers. Remote sensing data from Sentinel-1 and Sentinel-2 acquired in 2023 were used for the study. The Sentinel-1 data provided VV and VH polarization bands of SAR imagery, while Sentinel-2 supplied multi-spectral band data, enabling the classification of complex wetland features. The location and boundaries of the study area are illustrated in the Fig. 1.

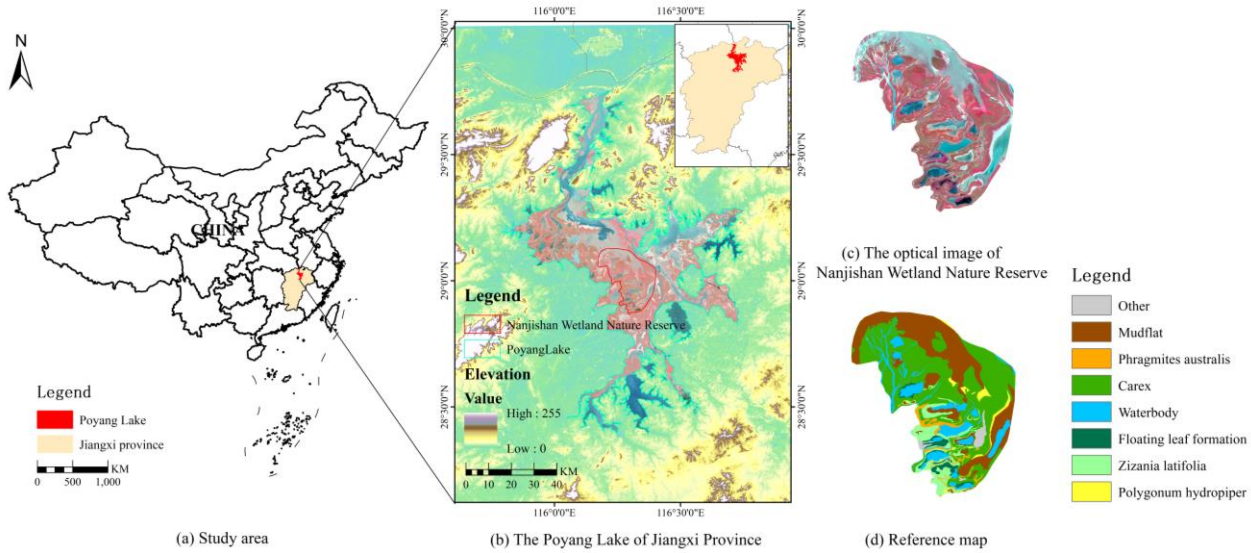


Fig. 1. (a) The location and extend of the study area, (b) a Digital Elevation Model (DEM) colorized by height, (c) optical image representation, and (d) the ground reference.

### B. Data set

In this study, Sentinel-1 and Sentinel-2 data from 2023 were used, combined with SNAP and ENVI software to complete pre-processing to construct multi-source remote sensing image dataset. Sentinel-1 data: After orbit correction, radiation calibration, specular filtering, terrain correction and decibel conversion, VV and VH polarization bands are extracted to improve data quality. Sentinel-2 data: Extract B2 (blue), B3 (green), B4 (red) and B8 (near infrared) bands and crop them to the study area to provide rich spectral information. The two kinds of data are registered and fused by ENVI to generate multi-modal feature data set. Because of the vast area of Poyang Lake wetland, the classification method is segmented, and the results are spliced into a complete regional map. Based on the "Chinese Wetland Classification System" [11] and GB/T14848-93 standard, combined with the actual situation of wetlands, the classification system is classified into: mudflat, reeds, carex, water, floating leaf plants, wild rice, Polygonum red and other categories. The reference map is generated by combining high-resolution images, field investigation and manual interpretation, as shown in Fig. 1.

## III. Method

In this study, a deep learning model named SOCANet is proposed for the classification of SAR and multispectral images. SOCANet features a two-branch architecture, incorporating a cross-modal feature fusion module and a multi-level feature extraction module. This design significantly enhances the classification accuracy of complex land cover types in the Nanjishan Wetland Nature Reserve.

### A. SOCANet architecture

The overall architecture of SOCANet is shown in Fig. 2, which mainly includes a cross-modal feature fusion module

and a multi-level feature extraction module. The feature extraction module is further divided into convolution branch and fusion linear attention branch.

First, SAR and multispectral images are input into the cross-modal feature fusion module respectively. Since SAR and optical images come from different sensors, there are significant semantic differences, and simple band superposition cannot effectively solve this problem. This module generates semantically aligned fusion feature maps through cross-modal feature fusion.

The fusion feature map then enters the multi-level feature extraction module, which consists of 12 stages to extract features step by step. In each stage, the network is divided into two branches, which receive the feature maps and feature vectors from the previous stage, to further enhance the information expression ability and realize the efficient classification of wetland complex ground objects.

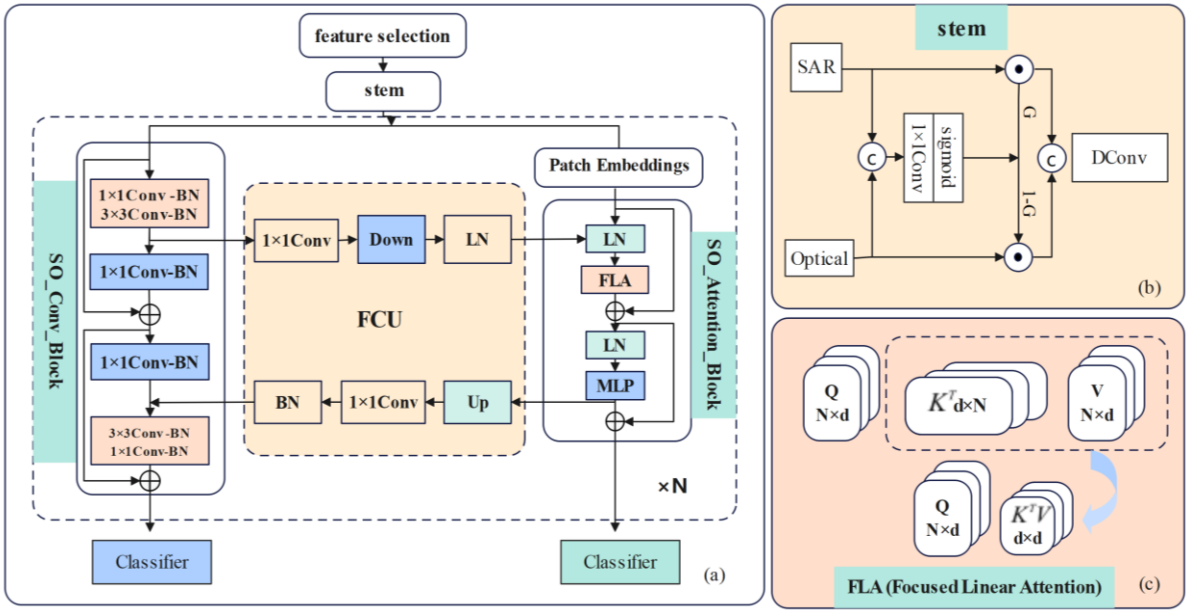


Fig. 2. SOCANet architecture, (a) is a monolithic framework; (b) is the stem component of the overall framework, consisting of GFM and DConv; (c) is the FLA (Focused Linear Attention) part of the overall framework

### *B. Cross-modal feature fusion module*

We employ the GFM proposed by Hosseinpour et al.[12] to align the features of optical images and SAR images. Within the GFM, the optical image and the SAR image are concatenated in the channel dimension, and then a 1 * 1 convolutional layer is utilized to establish cross-modal relationships and reduce the feature dimension to that of the module input. Subsequently, a sigmoid layer is adopted to obtain the weighting matrix G. The optical image and the SAR image are weighted by G and 1 - G through element-wise multiplication. Subsequently, the weighted features are concatenated in the channel dimension to yield the final output, namely the fused feature map.

The module effectively alleviates the semantic difference between SAR and optical images, and improves the expression ability of cross-modal features.

### *C. SO_Conv_Block*

The CNN branch adopts the feature pyramid structure, and the resolution of the feature map gradually increases with the depth of the network. The branch is divided into 4 stages, each consisting of multiple convolution blocks, each containing a bottleneck layer. The bottleneck structure is a reference to ResNet and consists of three convolution operations: a 1×1 lower projection convolution, a 3×3 spatial convolution, and a 1×1 upper projection convolution, while preserving the flow of information between input and output via residual joins. In addition, convolution kernels overlap and slide on the feature graph, effectively preserving local features. By extracting local features efficiently, the CNN branch provides refined local feature information to the transformer branch, thereby improving the feature expression capability of the overall network.

### *D. SO_Attention_Block*

This module is mainly composed of Focused Linear Attention (FLA), proposed by Han[13], which is a linear attention module that efficiently expresses information.

## IV. Results and Discussion

### *A. Implementation details*

All experiments were carried out using Python-3.9 and PyTorch. The evaluation is performed on the same device. The model was trained over 50 periods and the optimal weights were saved for prediction. Adam is used as an optimizer.

### *B. Comparison method*

We chose RMT, CoAtNet and Conformer, which are also CNN and self-attention mechanisms, to compare with our method, and the results are shown in the table below.

TABLE I. CONTRAST EXPERIMENT

| Method | OA(%) | Kappa | Iou(%) |
|---|---|---|---|
| RMT | 74.02% | 0.637 | 22.25% |
| CoAtNet | 95.28% | 0.935 | 76.55% |
| Conformer | 96.28% | 0.949 | 81.61% |
| SOCANet | 97.00% | 0.959 | 84.91% |

### C. *Accuracy evaluation*

In experiments with the Nanjishan Wetland Nature Reserve dataset using multi-source remote sensing data and various CNN and transformer based classifiers, our method demonstrates the best performance, as shown in Table I. The quantitative classification results shown in Table I show that our method has achieved significant improvement, achieving an overall accuracy (OA) of 97% and a Kappa coefficient of 0.959, which is much better than other classifiers based on CNN and transformer. Such as RMT CoAtNet, Conformer.

SOCANet has the best performance, with an overall accuracy rate (OA) of 97% and Kappa of 0.959, respectively. It can increase the contrast of the boundary between water and wetland vegetation, distinguish the strong difference between them, and is very effective in distinguishing complex ground objects in wetland environment.

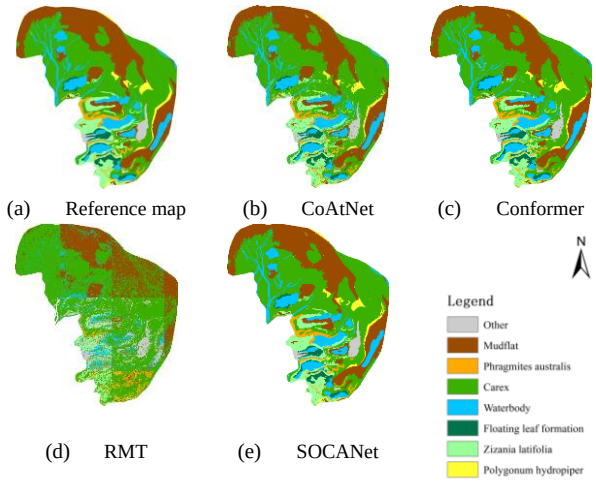


Fig. 3. (a) is the reference map, (b), (c), and (d) present a comparative analysis of experimental results corresponding to CoAtNet, Conformer, and RMT, respectively, (e) is our method SOCANet

## V. CONCLUSION

This study introduces a joint classification dynamic monitoring strategy based on CNN and Attention, and discusses the impact of feature alignment after fusion of optical and SAR remote sensing data and adding edge variable convolution to the model on the classification results of Nanjishan Wetland Nature Reserve. In the comparative experiment, our proposed method has the best performance, with an overall accuracy of 97% and Kappa of 0.959, respectively. It can increase the contrast of the boundary between water and wetland vegetation, distinguish the strong difference between them, and is very effective in distinguishing complex ground objects in wetland environment.

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